

Scorecard Update

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Overview

This markdown shows the data logging process of each round, and shows performance trends and metric averages

Set-Up Environment

Attach Packages

```
library(golf)
library(tidyverse)
library(DBI)
library(duckdb)
```

Record New Scorecard

Input the Scores Data

```
# required inputs to fill out the scorecard

round_course <- 'Randolph North'
round_date <- '2026-06-07'
round_teens <- 'white'

hole_scores <- c(4, 3, 6, 5, 4, 3, 3, 4, 5,
                 5, 3, 5, 3, 6, 4, 4, 5, 5)

FIRs <- c(1, rep(0, 5), 1, 0, 0,
          1, 0, 0, 1, rep(0, 5))

GIRs <- c(0, 1, 0, 0, 1, 0, 1, 0, 0,
          0, 1, 0, 1, 0, 0, 1, 0, 0)

putts_rec <- c(1, 1, 2, 2, 2, 1, 1, 1, 1,
               2, 2, 2, 1, 2, 2, 1, 2, 1)

chips_rec <- c(1, 0, 1, 1, 0, 1, 0, 2, 2,
```

```

        1, 0, 1, 0, 1, 1, 1, 1, 1)

penalties_rec <- c(rep(0, 18))

tee_clubs <- c('D', 'D', 'D', 'D', 'D', 'PW', '4', '7', 'D',
              'D', '7', 'D', 'D', 'D', '8', 'D', 'D', 'D')

index <- 11.0

```

Specify Course and Tees

```

# get the scorecard for the new round, ensuring hole-by-hole scores are filled
if ( length(hole_scores) > 0 ) {
  Card <- golf::get_course(course = round_course, date = round_date, tees = round_tees)
}

```

Get Scoring Metrics

```

# fill the scorecard for the new round, ensuring hole-by-hole scores are filled
if ( length(hole_scores) > 0 ) {
  Card <- golf::log_score(Scorecard = Card,
                        hole_by_hole = hole_scores,
                        GHIN = 10526424,
                        index = index,
                        FIR = FIRs,
                        GIR = GIRs,
                        putts = putts_rec,
                        chips = chips_rec,
                        penalties = penalties_rec,
                        tee_club = tee_clubs)
}

```

Update the db

Update the Players Table

```

con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'players' table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC"))
    dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0

```

```

    ) {

DBI::dbAppendTable(conn = con,
  name = 'players',
  value = players <- Card |>
    dplyr::select(GHIN, index, date) |>
    dplyr::distinct() |>
    dplyr::rename(handicap_index = index) |>
    dplyr::mutate(date = as.character(date))
  )
}

```

Update the Courses Table

```

con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'courses' table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC")) |>
  dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0
) {
DBI::dbAppendTable(conn = con,
  name = 'courses',
  value = course <- Card |>
    dplyr::select(course, tees, to_par, slope, course_rating, hole, yds, par, hole_handicap) |>
    dplyr::distinct() |>
    dplyr::rename(course_name = course) |>
    dplyr::mutate(hole = gsub(hole,
      pattern = 'hole_',
      replacement = '') |> as.numeric())
  )
}

```

Update the Rounds Table

```

con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'rounds' table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC")) |>
  dplyr::distinct(date) |>
  unlist() %>%

```

```

lubridate::as_date(.) |>
as.character() < round_date &

length(hole_scores) > 0
) {
DBI::dbAppendTable(conn = con,
  name = 'rounds',
  value = Card |>
    dplyr::select(-to_par, -slope, -course_rating, -yds, -par) |>
    dplyr::distinct() |>
    dplyr::rename(handicap_index = index) |>
    dplyr::rename(course_name = course) |>
    dplyr::mutate(date = as.character(date)) |>
    dplyr::mutate(hole = gsub(hole,
      pattern = 'hole_',
      replacement = '') |> as.numeric())
)
}

```

Get Club Metrics df

```

# create a dataframe, specifying the shape for tracked shots on each hole
# -> all strokes from off the green

club_metrics_df <- golf::get_tracked_shots_data_shape(round_date = round_date) |>
  dplyr::arrange(hole)

```

Annotate Club Metrics

```

# manually annotate from Garmin Golf App log
club_choice <- c(
  'D', 'GW', 'P',
  'D', 'GW',
  'D', '5', '5', 'PW',
  'D', '5', 'PW',
  'D', '9',
  'PW', 'PW',
  '4', 'PW',
  '7', 'GW', 'PW',
  'D', '5', '7', 'PW',

  'D', 'GW', 'LW',
  '7',
  'D', 'GW', 'P',
  'D', '4',
  'D', '4', '6', 'PW',
  '8', 'SW',
  'D', '3W', 'PW',
  'D', 'PW', 'SW',
  'D', '5', 'GW', 'LW'
)

```

```

)

dist_to_target <- c(
  270, 34, 8,
  270, 48,
  270, 205, 189, 8,
  270, 189, 16,
  270, 148,
  129, 30,
  220, 130,
  173, 25, 12,
  270, 209, 46, 21,

  270, 112, 18,
  185,
  270, 118, 9,
  270, 208,
  270, 220, 191, 13,
  168, 13,
  270, 233, 15,
  270, 148, 27,
  270, 206, 105, 11
)

yds <- c(
  310, 27, 8,
  308, 42,
  245, 71, 182, 6,
  241, 173, 13,
  281, 147,
  146, 24,
  224, 130,
  166, 12, 9,
  231, 188, 49, 23,

  288, 106, 13,
  173,
  267, 110, 9,
  256, 208,
  118, 100, 188, 13,
  159, 14,
  263, 242, 9,
  222, 130, 27,
  200, 101, 102, 7
)

lie_type <- c(
  'tee', 'fairway', 'fairway',
  'tee', 'rough',
  'tee', 'sand', 'fairway', 'fairway',
  'tee', 'rough', 'fairway',
  'tee', 'rough',
  'tee', 'rough',
  'tee', 'fairway',

```

```

'tee', 'rough', 'rough',
'tee', 'fairway', 'rough', 'rough',

'tee', 'fairway', 'sand',
'tee',
'tee', 'rough', 'fairway',
'tee', 'fairway',
'tee', 'rough', 'rough', 'fairway',
'tee', 'rough',
'tee', 'rough', 'rough',
'tee', 'fairway', 'sand',
'tee', 'fairway', 'fairway', 'sand'
)

target_status <- c(
  'yes', 'no', 'yes',
  'no', 'yes',
  'no', 'no', 'no', 'yes',
  'no', 'yes', 'yes',
  'no', 'yes',
  'no', 'yes',
  'yes', 'yes',
  'no', 'no', 'yes',
  'no', 'no', 'yes', 'yes',

  'yes', 'no', 'yes',
  'yes',
  'no', 'no', 'yes',
  'yes', 'yes',
  'no', 'no', 'no', 'yes',
  'no', 'yes',
  'no', 'no', 'yes',
  'no', 'no', 'yes',
  'no', 'no', 'no', 'yes'
)

location <- c(
  'on_target', 'short', 'on_target',
  'right', 'on_target',
  'right', 'short', 'short', 'on_target',
  'right', 'on_target', 'on_target',
  'left', 'on_target',
  'left', 'on_target',
  'on_target', 'on_target',
  'right', 'short', 'on_target',
  'right', 'right', 'on_target', 'on_target',

  'on_target', 'right', 'on_target',
  'on_target',
  'right', 'short', 'on_target',
  'on_target', 'on_target',
  'left', 'short', 'long', 'on_target',
  'short', 'on_target',

```

```

'right', 'long', 'on_target',
'right', 'left', 'on_target',
'right', 'short', 'left', 'on_target'
)

type_of_shot <- c(
  'tee', 'choked', 'chip',
  'tee', 'chip',
  'tee', 'fwbunker', 'full', 'chip',
  'tee', 'punch', 'chip',
  'tee', 'full',
  'tee', 'chip',
  'tee', 'full',
  'tee', 'chip', 'chip',
  'tee', 'full', 'chip', 'chip',

  'tee', 'choked', 'gsbunker',
  'tee',
  'tee', 'full', 'chip',
  'tee', 'full',
  'tee', 'punch', 'full', 'chip',
  'tee', 'chip',
  'tee', 'full', 'chip',
  'tee', 'full', 'gsbunker',
  'tee', 'full', 'choked', 'gsbunker'
)

```

Get Shot Metrics

```

con <- golf::get_db_connection()

# if the data is newer than the last round in the 'club_metrics' table, append
# the new annotations (expands from 18 rows by the number determined from the
# get_tracked_shots_data_shape function)

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM club_metrics ORDER BY date")) %>%
  dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0
) {
  length(club_choice)
  club_metrics <- golf::harmonize_club_metrics(club_metrics = club_metrics_df,
                                              club_choice = club_choice,
                                              distance_to_target = dist_to_target,
                                              distance_traveled = yds,
                                              lie_type = lie_type,
                                              target_status = target_status,
                                              location = location,
                                              type_of_shot = type_of_shot

```

```

    )
    club_metrics <- club_metrics |>
      dplyr::mutate(dplyr::across(c(dplyr::contains("yds")), ~as.numeric(.x)))
    head(club_metrics)
  }

club_metrics_df |> dplyr::select(tracked_shots) |> dplyr::mutate(sum(tracked_shots))

```

```

##   tracked_shots sum(tracked_shots)
## 1             3             50
## 2             2             50
## 3             4             50
## 4             3             50
## 5             2             50
## 6             2             50
## 7             2             50
## 8             3             50
## 9             4             50
## 10            3             50
## 11            1             50
## 12            3             50
## 13            2             50
## 14            4             50
## 15            2             50
## 16            3             50
## 17            3             50
## 18            4             50

```

Update the Club Metrics Table

```

con <- golf::get_db_connection()

# if the data is newer than the last round in the 'club_metrics' table, append
# the new data to the table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM club_metrics ORDER BY date")) |>
  dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0
) {
  DBI::dbAppendTable(conn = con,
    name = 'club_metrics',
    value = club_metrics |>
      dplyr::mutate(date = as.character(date))
  )
}

```



```
# clean the global environment
rm(list = ls()[which(grepl(ls(), pattern= 'con|round_course|round_date|index|real_db')==F)])
```

Summarize Metrics

Gather and Format

Gather and format from the database

```
con <- golf::get_db_connection()

# get the hole-by-hole scores from each round in the database

# join the hole-by-hole scores from the 'rounds' table to the 'courses' table
# need each hole's par rating
# need each course's course_rating from the 'courses' table

# join the 'rounds' and 'courses' tables to the 'players' table
# need handicap_index from the 'players' table

scores <- DBI::dbGetQuery(conn = con, statement = paste0(
  "SELECT DISTINCT r.*, c.par, c.course_rating FROM rounds r
  INNER JOIN courses c
  ON c.tees = r.tees
  AND c.course_name = r.course_name
  AND c.hole = r.hole
  INNER JOIN players p
  ON r.GHIN = p.GHIN
  AND r.handicap_index = p.handicap_index
  AND r.date = p.date;"
)) |>
  dplyr::mutate(date = lubridate::as_date(date)) |> # convert strings to date's
  dplyr::relocate(par, .after = hole) |>
  dplyr::relocate(course_rating, .after = tees) |>
  dplyr::group_by(date) |>
  dplyr::arrange(desc(date), hole) |>
  dplyr::ungroup()
```

```
con <- golf::get_db_connection()

# get stroke metrics from the 'club_metrics' table

stroke_quality <- DBI::dbGetQuery(conn = con, statement = paste0(
  "SELECT DISTINCT * FROM club_metrics;"
)) |>
  dplyr::mutate(date = lubridate::as_date(date)) |> # convert strings to date's
  dplyr::group_by(date) |>
  dplyr::arrange(desc(date), hole, stroke) |>
  dplyr::ungroup()
```

Compute Metrics

Compute standard metrics

```
# define metrics
scores_sum <- scores |>
  dplyr::mutate(date_course = paste0(date,
                                     '\n',
                                     course_name, '\n',
                                     handicap_index),
               chips = dplyr::case_when(
                 grepl(date, pattern = '07-13') ~ NA_real_, TRUE ~ chips),
               `chips+putts` = chips+putts,
               FIR_opps = dplyr::case_when(par > 3 ~ 1, TRUE ~ NA),
               `Iron FIRs` = dplyr::case_when(par > 3 &
                                               grepl(tee_club, pattern = '4|5') &
                                               FIR == 1 ~ 1,
                                               TRUE ~ 0),
               `Iron FIR opps` = dplyr::case_when(par > 3 &
                                                  grepl(tee_club, pattern = '4|5') ~ 1,
                                                  TRUE ~ 0),
               `Driver FIRs` = dplyr::case_when(par > 3 &
                                                  tee_club == 'D' &
                                                  FIR == 1 ~ 1,
                                                  TRUE ~ 0),
               `Driver FIR opps` = dplyr::case_when(par > 3 &
                                                  tee_club == 'D' ~ 1,
                                                  TRUE ~ 0),
               `Par 3 GIRs` = dplyr::case_when(par == 3 &
                                                  GIR == 1 ~ 1,
                                                  TRUE ~ 0),
               greenie_putts = dplyr::case_when(GIR == 1 ~ putts, TRUE ~ NA_real_),
               updown_conv = dplyr::case_when(GIR == 0 &
                                               par == gross ~ 1,
                                               TRUE ~ 0),
               updown_opps = dplyr::case_when(GIR == 0 &
                                               chips > 0 ~ 1,
                                               TRUE ~ 0)
  ) |>
  dplyr::rename(`Handicap Index` = handicap_index)

# summarize by round
scores_sum <- scores_sum |>
  dplyr::group_by(date, date_course, course_rating, `Handicap Index`) |>
  dplyr::summarize(
    FIRs = sum(FIR, na.rm = F),
    `Iron FIRs` = sum(`Iron FIRs`, na.rm = T),
    `Iron FIR%` = round(((sum(`Iron FIRs`, na.rm = T)/sum(`Iron FIR opps`, na.rm = T))*100, 1),
    `Driver FIRs` = sum(`Driver FIRs`, na.rm = T),
    `Driver FIR%` = round(((sum(`Driver FIRs`, na.rm = T)/sum(`Driver FIR opps`, na.rm = T))*100, 1),
    `FIR%` = round((sum(FIRs, na.rm = T)/sum(FIR_opps, na.rm = T))*100, 1),
    GIRs = sum(GIR, na.rm = F),
    `Par 3 GIRs` = sum(`Par 3 GIRs`, na.rm = T),
    `GIR%` = round((sum(GIRs, na.rm = T)/18)*100, 1),
```

```

    putts = sum(putts, na.rm = F),
    `Avg GIR putts` = round(mean(greenie_putts, na.rm = T), 2),
    chips = sum(chips, na.rm = F),
    `chips+putts` = sum(`chips+putts`, na.rm = F),
    `UpDown%` = round(((sum(updown_conv, na.rm = T)/sum(updown_opps, na.rm = T))*100), 1),
    pars = sum(is_gross_par, na.rm = F),
    birdies = sum(is_gross_birdie, na.rm = F),
    bogies = sum(is_gross_bogey, na.rm = F),
    `doubles+` = sum(is_gross_bogey_worse, na.rm = F),
    penalties = sum(penalties, na.rm = T),
    `Gross Score` = mean(tot_gross, na.rm = F),
    `Net Score` = mean(tot_net, na.rm = F)) %>%
dplyr::mutate(dplyr::across(c(`Iron FIRs`, `Driver FIRs`,
                             `Iron FIR%`, `Driver FIR%`, `FIR%`),
                           ~dplyr::if_else(is.na(FIRs), NA, .)),

  dplyr::across(c(`Par 3 GIRs`, `Avg GIR putts`,
                  `UpDown%`, `GIR%`),
                ~dplyr::if_else(is.na(GIRs), NA, .)),

  `chips+putts` = dplyr::case_when(is.na(chips) ~ NA_real_,
                                  TRUE ~ `chips+putts`) %>%

dplyr::mutate(

  `UpAndDown%` = dplyr::case_when(

    grepl(date, pattern = '07-13|09-21') ~ NA, TRUE ~ `UpDown%`),

  `Iron FIR%` = dplyr::case_when(`Iron FIR%` == NaN ~ 0.0, TRUE ~ `Iron FIR%`))

```

```

head(scores_sum |>
  dplyr::arrange(desc(date)))

```

```

## # A tibble: 6 x 26
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating `Handicap Index`  FIRs `Iron FIRs` `Iron
##   <date>    <chr>          <dbl>          <dbl> <int>      <dbl>
## 1 2026-06-07 "2026-06-07\nRandolph North\n11"      69          11      4        1
## 2 2026-06-07 "2026-06-07\nRandolph North\n11"     69.8        11      4        1
## 3 2026-05-31 "2026-05-31\nFred Enke\n11"      68.6        11      3        1
## 4 2026-05-24 "2026-05-24\nQuarry Pines\n11"     68.8        11      9        7
## 5 2026-05-17 "2026-05-17\nRandolph North\n11"     70.4        11      2        0
## 6 2026-04-26 "2026-04-26\nDell Urich\n11"      68.5        11      5        0

```

View Metrics

Separate and view metrics

Scoring Metrics Scores and Handicap

```
scoring_metrics <- scores_sum |>
  dplyr::select(`Handicap Index`, `Gross Score`, `Net Score`)
head(scoring_metrics |>
  dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 6
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating `Handicap Index` `Gross Score` `Net Score`
##   <date>    <chr>          <dbl>          <dbl>          <dbl>          <dbl>
## 1 2026-06-07 "2026-06-07\nRandolph North\n11"      69              11              77              0
## 2 2026-06-07 "2026-06-07\nRandolph North\n11"     69.8             11              77              0
## 3 2026-05-31 "2026-05-31\nFred Enke\n11"        68.6             11              90              1
## 4 2026-05-24 "2026-05-24\nQuarry Pines\n11"    68.8             11              86              1
## 5 2026-05-17 "2026-05-17\nRandolph North\n11"    70.4             11              85              1
## 6 2026-04-26 "2026-04-26\nDell Urich\n11"      68.5             11              92              0
```

Stroke Metrics In golf, every hole has a **par**— an average number of strokes taken to get the ball in the hole

There are (almost always), three different **pars** on every course:

- par 3s
- par 4s
- par 5s

A shorthand to determine how a golfer performed across holes is to rate how many strokes *relative to par* they were (-1, -2, +1, +2, etc)

These numbers are given names:

- 0 = par
- -2 = eagle
- -1 = birdie
- +1 = bogey
- +2 = double bogey

I quantify these below:

```
stroke_metrics <- scores_sum |>
  dplyr::select(`doubles+`, bogies, pars, birdies)
head(stroke_metrics |>
  dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 7
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating `doubles+` bogies  pars birdies
##   <date>    <chr>          <dbl>          <int>  <int> <int>  <int>
## 1 2026-06-07 "2026-06-07\nRandolph North\n11"      69              1      8      5      3
## 2 2026-06-07 "2026-06-07\nRandolph North\n11"     69.8             1      8      5      3
## 3 2026-05-31 "2026-05-31\nFred Enke\n11"        68.6             5      7      5      1
## 4 2026-05-24 "2026-05-24\nQuarry Pines\n11"    68.8             4      6      8      0
## 5 2026-05-17 "2026-05-17\nRandolph North\n11"    70.4             3      8      6      1
## 6 2026-04-26 "2026-04-26\nDell Urich\n11"      68.5             6      7      4      1
```

Around-the-Green Metrics Chips: + strokes around the green taken to get onto the green

Putts: + strokes taken with the putter on the green

Avg GIR putts: + Average # of putts on holes where the ball was hit on to the green within 2 strokes of par (green in regulation, GIR)

UpDown% (aka Scramble%): + # of holes without a GIR but par was made / # of holes without a GIR

```
atg_metrics <- scores_sum |>
  dplyr::select(chips, `chips+putts`, `UpAndDown%`, putts, `Avg GIR putts`)
head(atg_metrics |>
  dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 8
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating chips `chips+putts` `UpAndDown%` putts
##   <date>    <chr>                <dbl> <dbl>      <dbl>      <dbl> <int>
## 1 2026-06-07 "2026-06-07\nRandolph North\n11"      69      15        42        25      27
## 2 2026-06-07 "2026-06-07\nRandolph North\n11"     69.8    15        42        25      27
## 3 2026-05-31 "2026-05-31\nFred Enke\n11"      68.6    13        47        20      34
## 4 2026-05-24 "2026-05-24\nQuarry Pines\n11"      68.8    17        53         0      36
## 5 2026-05-17 "2026-05-17\nRandolph North\n11"     70.4    17        46       31.2     29
## 6 2026-04-26 "2026-04-26\nDell Urich\n11"      68.5    19        55       15.4     36
```

Ball Striking Metrics Approach and tee accuracy

GIR (green in regulation): + hole where the ball was hit on to the green within 2 strokes of par

FIR (fairway in regulation): + hole where the ball was hit on to the fairway from the tee box (only on par 4's and par 5's)

Iron FIR: + hole where an iron was used off the tee for a FIR

Driver FIR: + hole where driver was used off the tee for a FIR

```
ball_striking_metrics <- scores_sum |>
  dplyr::select(GIRs, `GIR%`, `Par 3 GIRs`,
    FIRs, `FIR%`, `Iron FIRs`, `Iron FIR%`,
    `Driver FIRs`, `Driver FIR%`)
head(ball_striking_metrics |>
  dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 12
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating GIRs `GIR%` `Par 3 GIRs` FIRs `FIR%`
##   <date>    <chr>                <dbl> <int> <dbl>      <dbl> <int> <dbl>
## 1 2026-06-07 "2026-06-07\nRandolph North\n11"      69      6  33.3         1     4  28.6
## 2 2026-06-07 "2026-06-07\nRandolph North\n11"     69.8     6  33.3         1     4  28.6
## 3 2026-05-31 "2026-05-31\nFred Enke\n11"      68.6     5  27.8         2     3  21.4
## 4 2026-05-24 "2026-05-24\nQuarry Pines\n11"      68.8    10  55.6         2     9  69.2
## 5 2026-05-17 "2026-05-17\nRandolph North\n11"     70.4     2  11.1         1     2  14.3
## 6 2026-04-26 "2026-04-26\nDell Urich\n11"      68.5     3  16.7         0     5  38.5
```

Club Metrics Yardage and accuracy for each club

```
## # A tibble: 6 x 6
## # Groups:   date [1]
##   date      club      n rd_min_yds_to_target rd_min_yds_traveled rd_min_yd_diff
##   <date>    <chr> <int>          <dbl>          <dbl>          <dbl>
## 1 2026-06-07 3W      1           233            242            -9
## 2 2026-06-07 4        3           208            100            -4
## 3 2026-06-07 5        5           189             71             7
## 4 2026-06-07 6        1           191            188             3
## 5 2026-06-07 7        3            46             49            -3
## 6 2026-06-07 8        1           168            159             9
```

```
## # A tibble: 6 x 6
## # Groups:   date [1]
##   date      club      n rd_max_yds_to_target rd_max_yds_traveled rd_max_yd_diff
##   <date>    <chr> <int>          <dbl>          <dbl>          <dbl>
## 1 2026-06-07 3W      1           233            242            -9
## 2 2026-06-07 4        3           220            224            120
## 3 2026-06-07 5        5           209            188            134
## 4 2026-06-07 6        1           191            188             3
## 5 2026-06-07 7        3           185            173            12
## 6 2026-06-07 8        1           168            159             9
```

```
## # A tibble: 6 x 10
## # Groups:   date [1]
##   date      club `rd club strokes` miss_direction `rd club miss dir` rd_avg_yds_to_target rd_avg_yd_diff
##   <date>    <chr>          <int> <chr>                  <int>          <dbl>
## 1 2026-06-07 3W      1 long                    1           233
## 2 2026-06-07 4        2 on_target                2           214
## 3 2026-06-07 5        3 right                    1           201.
## 4 2026-06-07 5        3 short                    2           201.
## 5 2026-06-07 6        1 long                    1           191
## 6 2026-06-07 7        2 on_target                1           179
```

Plot Metric Summaries

Scoring Metrics

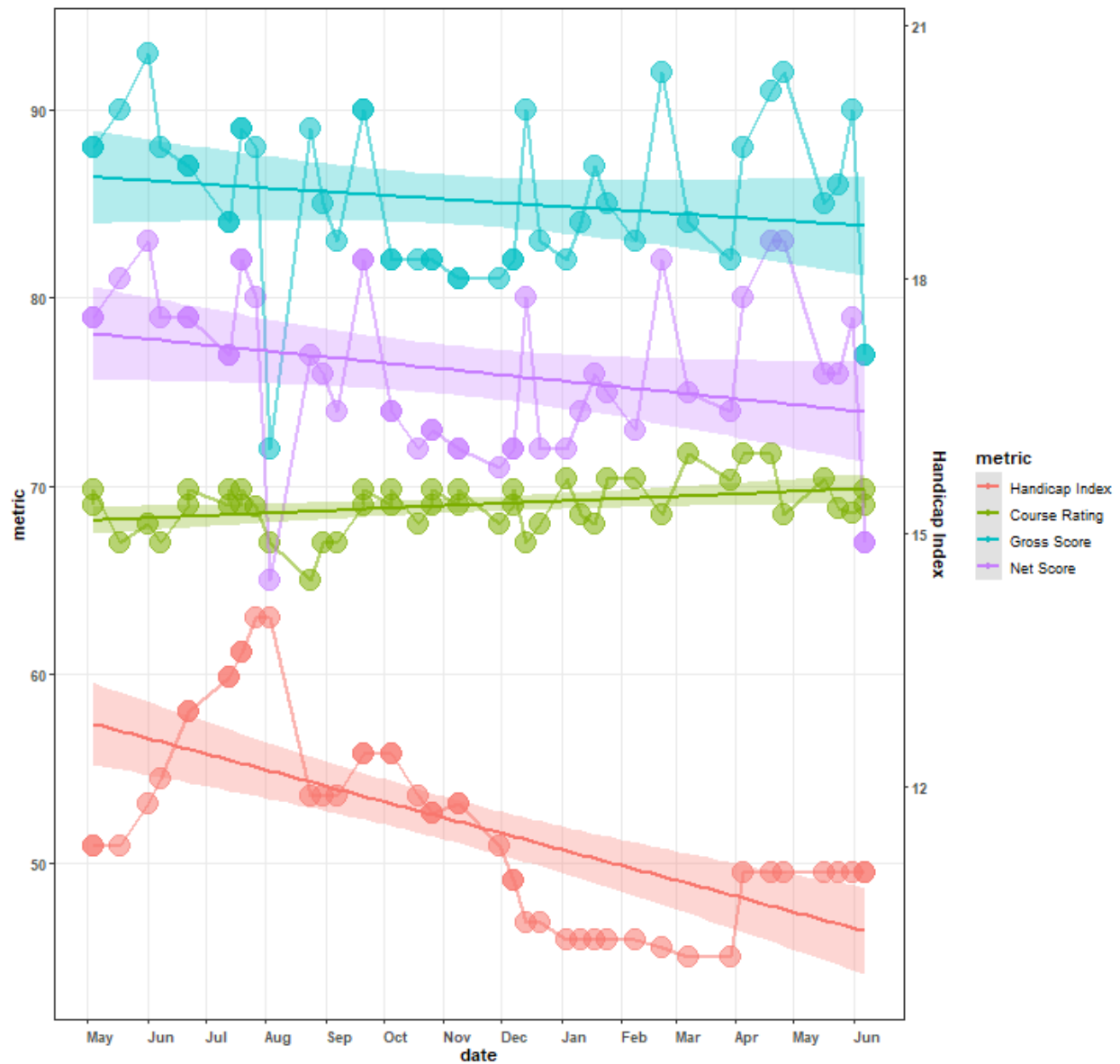
```
scoring_metrics |>
  dplyr::mutate(course_rating = dplyr::case_when(grepl(date_course, pattern = 'Sewailo') ~ 68.9, TRUE ~
    `Handicap Index` = `Handicap Index`*4.5) |>
  dplyr::rename(`Course Rating` = course_rating) |>
  dplyr::group_by(date, date_course, `Handicap Index`) |>
  tidyr::pivot_longer(cols = c(`Course Rating`, `Net Score`), names_to = 'metric', values_to = 'value', v
  ggplot2::ggplot(aes(x = date_course, y = value)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1,
    color = factor(metric,
      levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
    fill = factor(metric,
      levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
```

```

ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1,
                      color = factor(metric,
                                     levels = c('Handicap Index', 'Course Rating', 'Gross Score', ' '),
                                     fill = factor(metric,
                                     levels = c('Handicap Index', 'Course Rating', 'Gross Score', ' '),
                                     alpha = 0.3, method = 'lm')) +
ggplot2::geom_smooth(aes(x = date, y = value, group = metric,
                        color = factor(metric,
                                     levels = c('Handicap Index', 'Course Rating', 'Gross Score', ' '),
                                     fill = factor(metric,
                                     levels = c('Handicap Index', 'Course Rating', 'Gross Score', ' '),
                                     alpha = 0.3, method = 'lm')) +
ggplot2::scale_y_continuous(sec.axis = ggplot2::sec_axis(~./4.5, name = 'Handicap Index'))+
ggplot2::labs(title = paste0('Performance Over Time\n',
                             scores |>
                               dplyr::distinct(date) |>
                               dplyr::last() |>
                               unlist() %>% lubridate::as_date(.) |>
                               as.character(),
                             ' - ',
                             scores |>
                               dplyr::distinct(date) |>
                               dplyr::first() |>
                               unlist() %>% lubridate::as_date(.) |>
                               as.character()),
              x = 'date',
              y = 'metric',
              color = 'metric'
            ) +
ggplot2::guides(alpha = 'none', size = 'none', fill = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(panel.grid.minor = ggplot2::element_blank(),
              axis.title = ggplot2::element_text(face = 'bold'),
              axis.text.y = ggplot2::element_text(face = 'bold'),
              axis.text.x = ggplot2::element_text(face = 'bold', angle = 0, hjust = 0, vjust = 0),
              title = ggplot2::element_text(face = 'bold')
            ) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b')

```

Performance Over Time 2025-05-04 - 2026-06-07



Stroke Metrics

```
stroke_metrics |>
  tidyr::pivot_longer(cols = c(`doubles+`:birdies), names_to = 'metric', values_to = 'value', values_drop_na = T) +
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Hole-by-Hole Scores Over Time\n',
                                scores |>
                                  dplyr::distinct(date) |>
                                    dplyr::last() |>
```

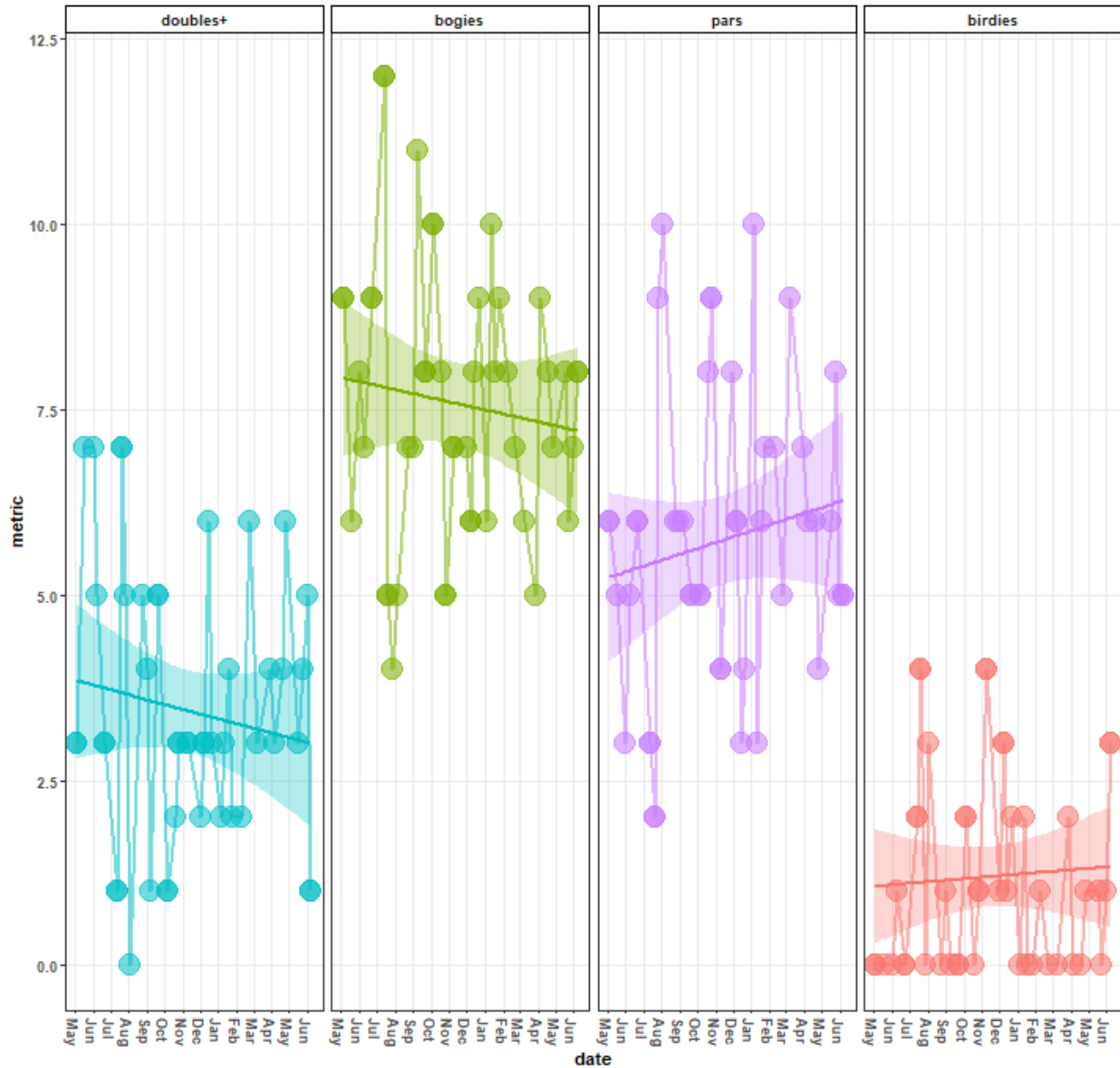


```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
    panel.grid.minor = ggplot2::element_blank(),
    axis.title = ggplot2::element_text(face = 'bold'),
    axis.text.y = ggplot2::element_text(face = 'bold'),
    axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
    title = ggplot2::element_text(face = 'bold'),
    strip.text.x.top = ggplot2::element_text(face = 'bold'),
    strip.background = ggplot2::element_rect(color = 'black',
        fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric, levels = c('doubles+', 'bogies', 'pars', 'birdies')), nrow = 1)

```

Hole-by-Hole Scores Over Time 2025-05-04 - 2026-06-07



Around the Green Metrics

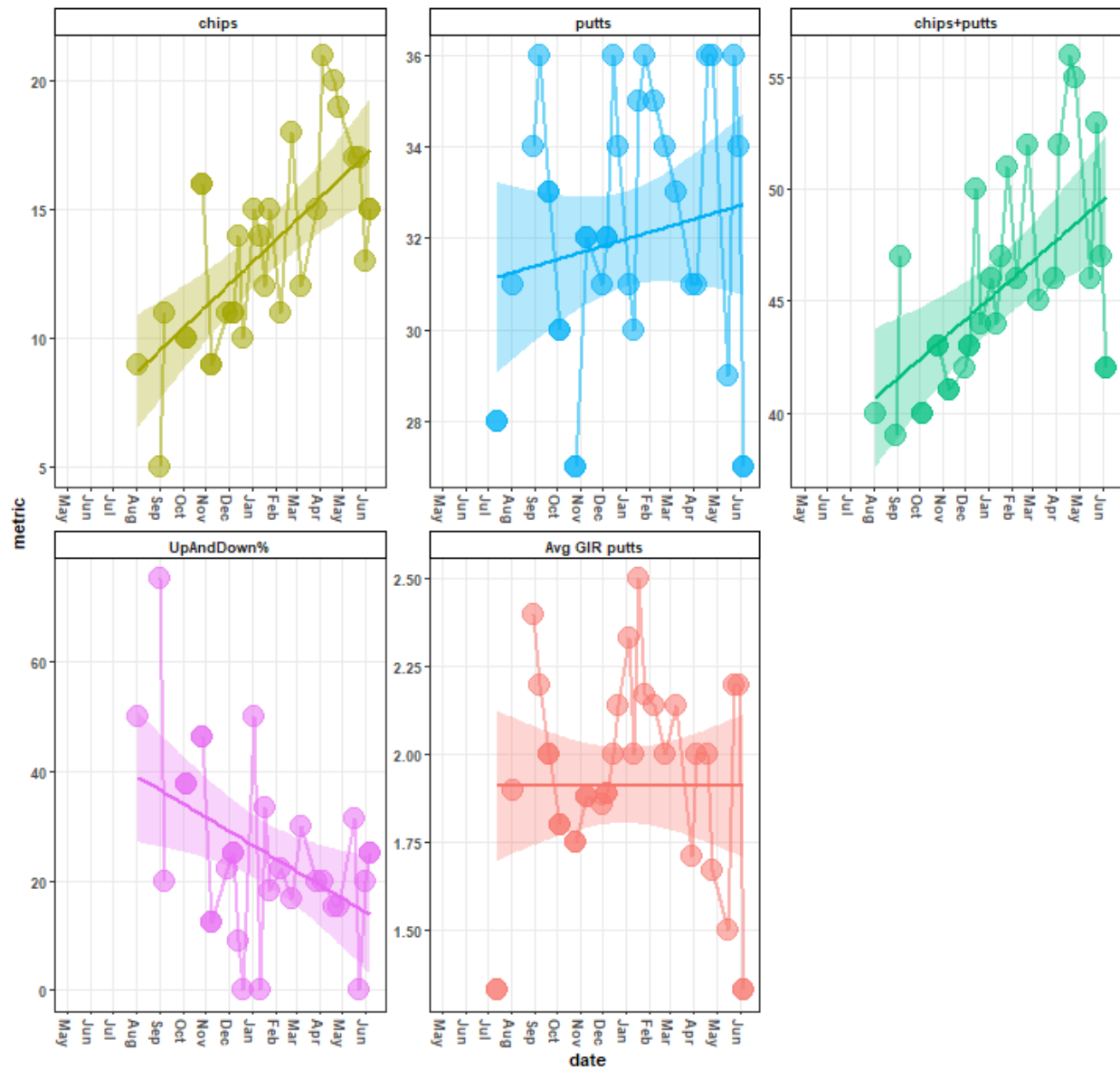
```
atg_metrics |>
  tidyr::pivot_longer(cols = c(chips:`Avg GIR putts`), names_to = 'metric', values_to = 'value', values_from = NA) +
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Around the Green Metrics Over Time\n',
                                scores |>
                                  dplyr::distinct(date) |>
                                    dplyr::last() |>
```

```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
               panel.grid.minor = ggplot2::element_blank(),
               axis.title = ggplot2::element_text(face = 'bold'),
               axis.text.y = ggplot2::element_text(face = 'bold'),
               axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
               title = ggplot2::element_text(face = 'bold'),
               strip.text.x.top = ggplot2::element_text(face = 'bold'),
               strip.background = ggplot2::element_rect(color = 'black',
                                                         fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric, levels = c('chips', 'putts', 'chips+putts', 'UpAndDown%', 'Avg GII

```

Around the Green Metrics Over Time 2025-05-04 - 2026-06-07



Ball Striking Metrics

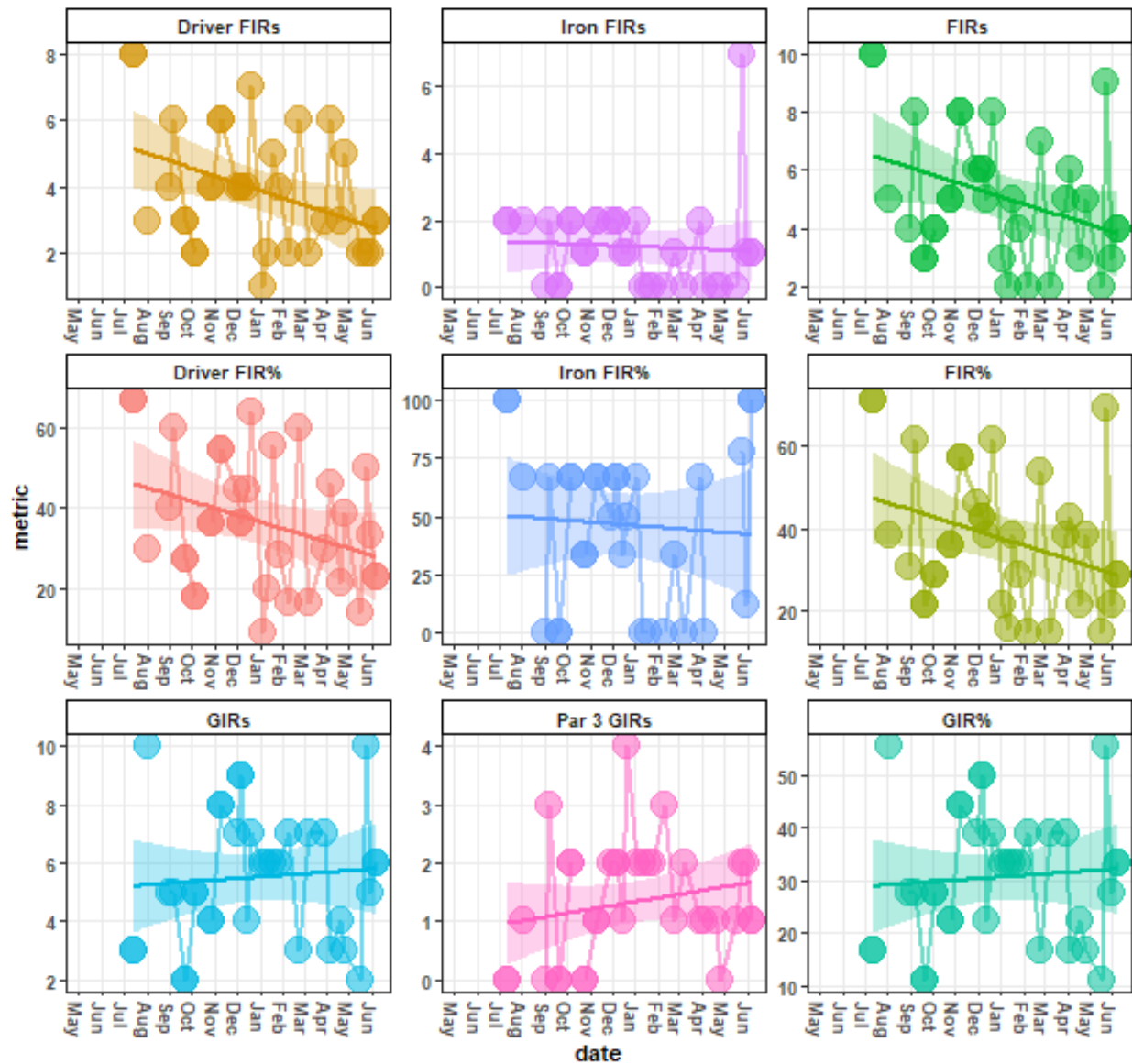
```
ball_striking_metrics |>
  tidyr::pivot_longer(cols = c(GIRs:`Driver FIR%`), names_to = 'metric', values_to = 'value', values_drop_na = T) +
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Ball Striking Over Time\n',
                                scores |>
                                  dplyr::distinct(date) |>
                                    dplyr::last() |>
```

```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
               panel.grid.minor = ggplot2::element_blank(),
               axis.title = ggplot2::element_text(face = 'bold'),
               axis.text.y = ggplot2::element_text(face = 'bold'),
               axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
               title = ggplot2::element_text(face = 'bold'),
               strip.text.x.top = ggplot2::element_text(face = 'bold'),
               strip.background = ggplot2::element_rect(color = 'black',
                                                         fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric, levels = c('Driver FIRs', 'Iron FIRs', 'FIRs',
                                              'Driver FIR%', 'Iron FIR%', 'FIR%',
                                              'GIRs', 'Par 3 GIRs', 'GIR%')), scales = 'free')

```

Ball Striking Over Time 2025-05-04 - 2026-06-07



Stroke Quality Metrics

```
stroke_quality_min |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(c(date:n), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains('yd')), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(
    grepl(metric, pattern = 'target') ~ 'min. target distance',
    grepl(metric, pattern = 'traveled') ~ 'min. shot distance',
    grepl(metric, pattern = 'diff') ~ 'min. \ntarget dist. - shot d

  dplyr::mutate(date = lubridate::as_date(date),
    club = as.character(club),
```

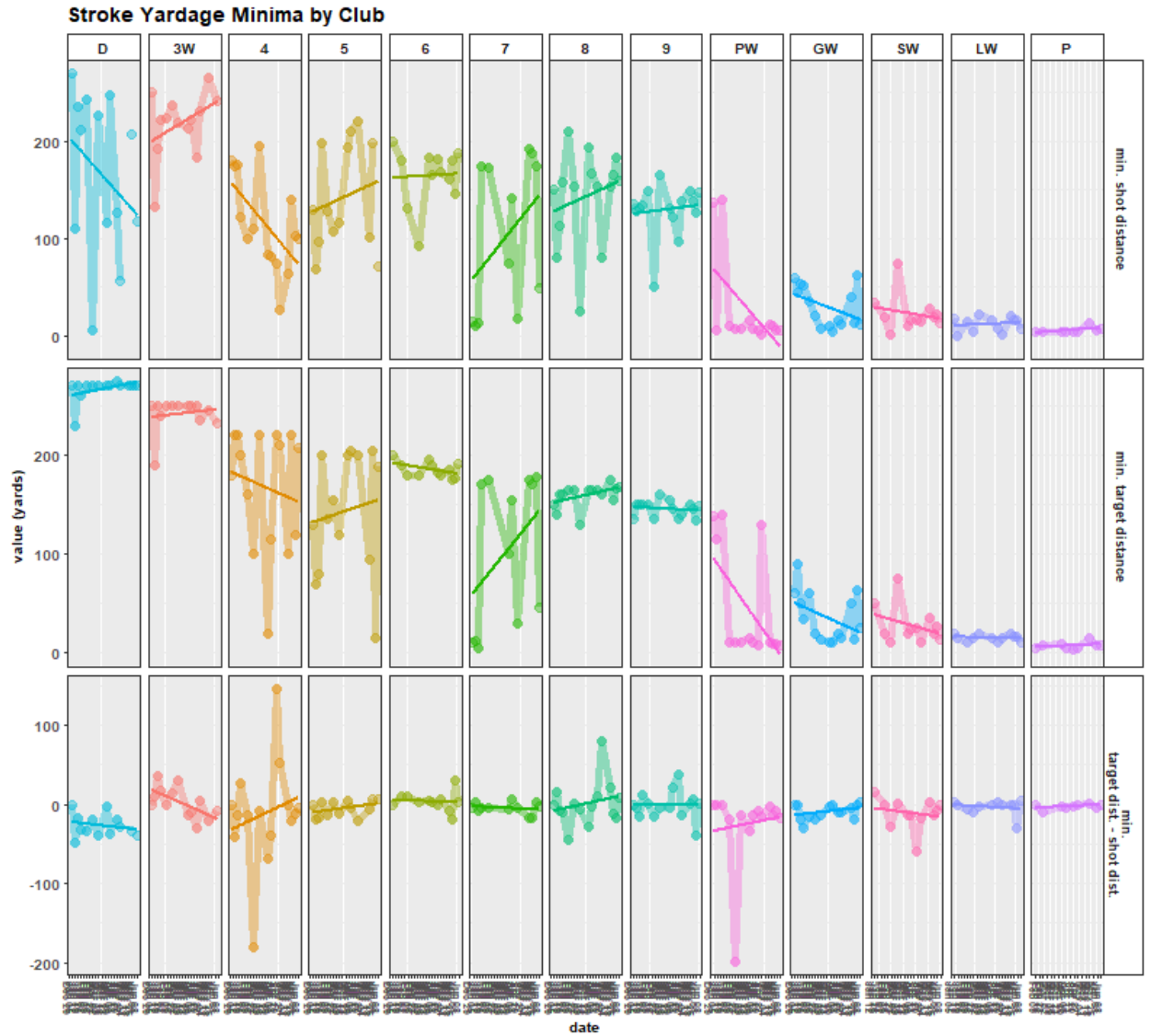
```

      n = as.integer(n)) |>
ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
ggplot2::geom_point(alpha = 0.4, size = 3) +
ggplot2::geom_line(alpha = 0.4, size = 2) +
ggplot2::geom_smooth(method = 'lm', se = F) +
ggplot2::labs(title = 'Stroke Yardage Minima by Club', x = 'date', y = 'value (yards)', fill = 'club')
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, vjust = 1),
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(

  rows = vars(metric),

  cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                     '6', '7', '8', '9',
                                     'PW', 'GW', 'SW', 'LW', 'P'))),
          scales = 'free')

```



Minima

```
stroke_quality_max |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(c(date:n), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains('yd')), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'max. target distance',
                                          grepl(metric, pattern = 'traveled') ~ 'max. shot distance',
                                          grepl(metric, pattern = 'diff') ~ 'max.\ntarget dist. - shot dist.'))
  dplyr::mutate(date = lubridate::as_date(date),
                club = as.character(club),
                n = as.integer(n)) |>
  ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
  ggplot2::geom_point(alpha = 0.4, size = 3) +
  ggplot2::geom_line(alpha = 0.4, size = 2) +
  ggplot2::geom_smooth(method = 'lm', se = F) +
  ggplot2::labs(title = 'Stroke Yardage Maxima by Club', x = 'date', y = 'value (yards)', fill = 'club')
```



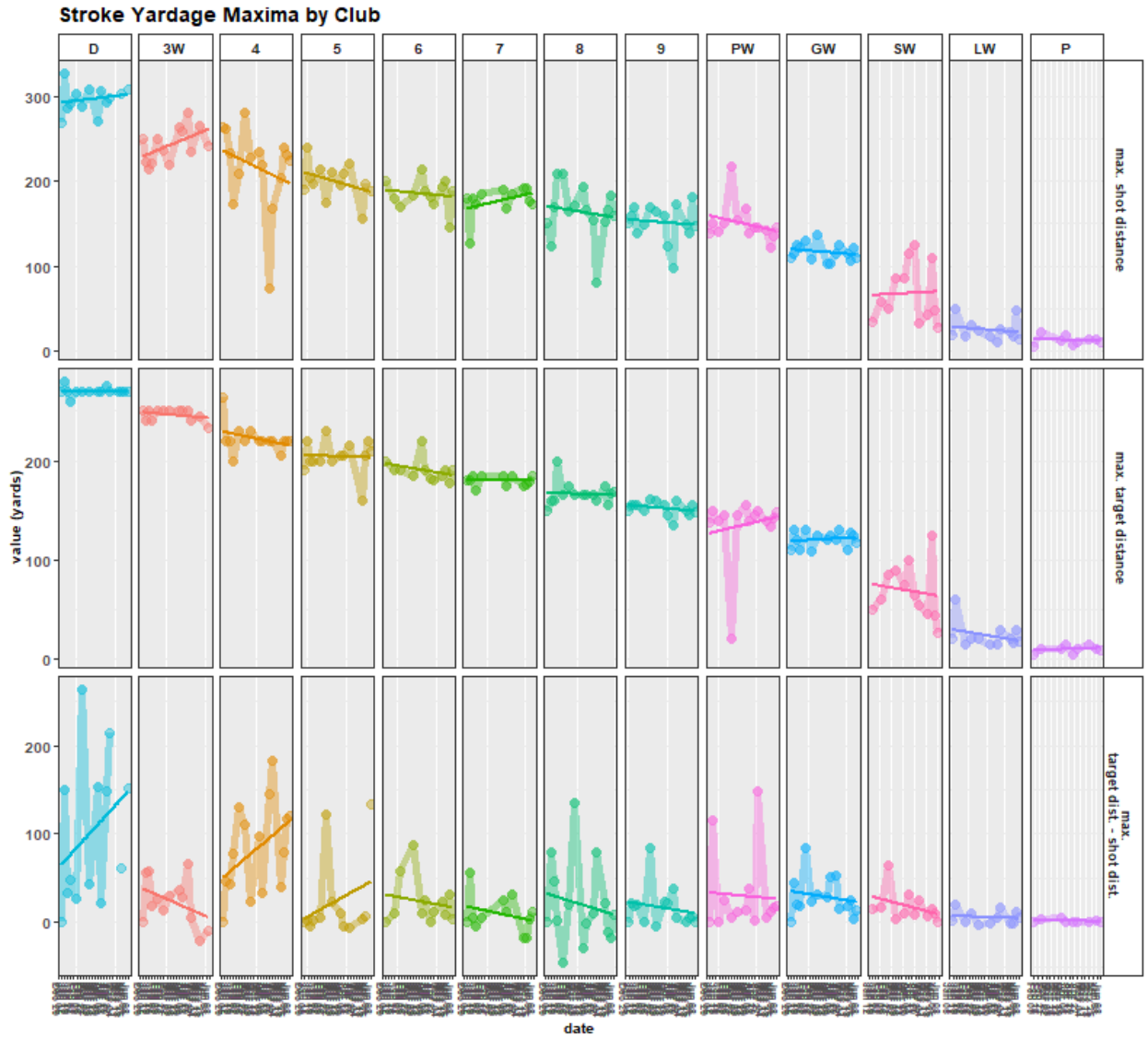
```

ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, vjust = 1),
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(

  rows = vars(metric),

  cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                     '6', '7', '8', '9',
                                     'PW', 'GW', 'SW', 'LW', 'P'))),
          scales = 'free')

```



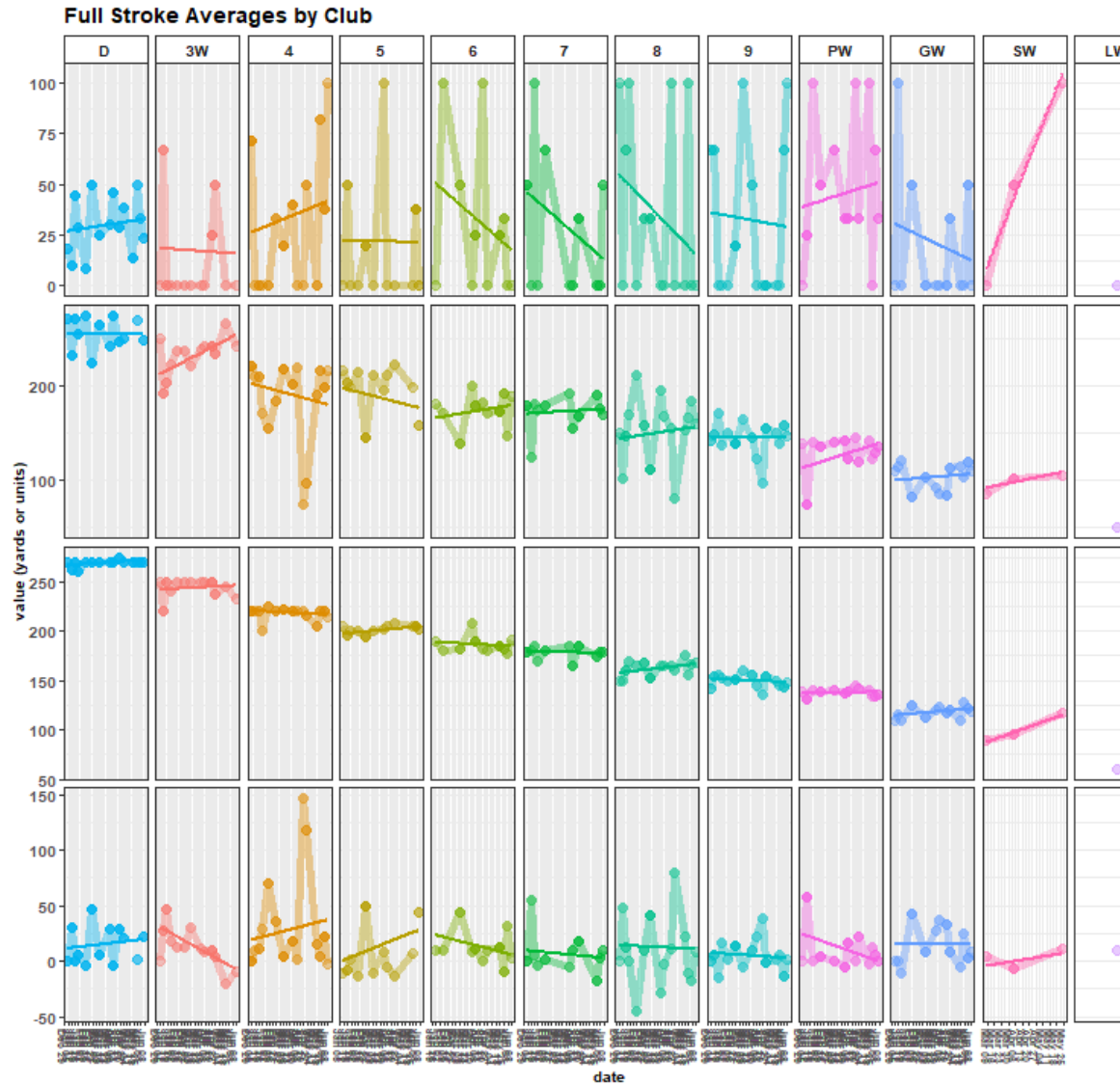
Maxima

```
full_stroke_quality_avg |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(dplyr::everything(), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains("rd")), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'avg. target distance',
    grepl(metric, pattern = 'traveled') ~ 'avg. shot distance',
    grepl(metric, pattern = 'diff') ~ 'avg. target dist. - shot dist.',
    # grepl(metric, pattern = 'accuracy') ~ 'avg. accuracy',
    grepl(metric, pattern = 'strokes') ~ 'n strokes',
    grepl(metric, pattern = 'dir') ~ 'strokes w/club\nin round\nby',
    grepl(metric, pattern = 'acc') ~ '% accuracy')) |>
  dplyr::filter(!grepl(metric, pattern = 'strokes')) |>
  dplyr::mutate(date = lubridate::as_date(date),
    club = as.character(club),
    vals = as.numeric(vals))
```

```

    ) |>
  ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
  ggplot2::geom_point(alpha = 0.4, size = 3) +
  ggplot2::geom_line(alpha = 0.4, size = 2) +
  ggplot2::geom_smooth(method = 'lm', se = F) +
  ggplot2::labs(title = 'Full Stroke Averages by Club', x = 'date', y = 'value (yards or units)', fill = club) +
  ggplot2::guides(alpha = 'none', size = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(
    title = ggplot2::element_text(face = 'bold', size = 12),
    axis.title = ggplot2::element_text(face = 'bold', size = 10),
    axis.text = ggplot2::element_text(face = 'bold', size = 10),
    axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, vjust = 1),
    legend.position = 'none',
    strip.background = ggplot2::element_rect(fill = 'white'),
    strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
    strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
  ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
  ggplot2::facet_grid(
    rows = vars(metric),
    cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                       '6', '7', '8', '9',
                                       'PW', 'GW', 'SW', 'LW', 'P'))),
    scales = 'free')

```



Full Stroke Average

Main Metrics

```
scores_sum |>
  dplyr::mutate(course_rating = dplyr::case_when(grepl(date_course, pattern = 'Sewailo') ~ 68.9, TRUE ~
  tidyr::pivot_longer(cols = c(`Handicap Index`, `Gross Score`, `Net Score`,
    `doubles+`, bogies, pars, birdies, chips, putts,
    `UpDown%`, `Avg GIR putts`, `GIR%`, `Par 3 GIRs`,
    `Iron FIR%`, `Driver FIR%`), names_to = 'metric', values_to = 'value', v
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Performance Over Time\n',
    scores |>
      dplyr::distinct(date) |>
      dplyr::last() |>
```

```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
  panel.grid.minor = ggplot2::element_blank(),
  axis.title = ggplot2::element_text(face = 'bold'),
  axis.text.y = ggplot2::element_text(face = 'bold'),
  axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
  title = ggplot2::element_text(face = 'bold'),
  strip.text.x.top = ggplot2::element_text(face = 'bold'),
  strip.background = ggplot2::element_rect(color = 'black',
    fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric,
  levels = c('Handicap Index', 'Gross Score', 'Net Score',
    'doubles+', 'bogies', 'pars', 'birdies',
    'chips', 'putts', 'UpDown%', 'Avg GIR putts',
    'GIR%', 'Par 3 GIRs', 'Iron FIR%', 'Driver FIR%')),
  scales = 'free',
  ncol = 4)

```

Performance Over Time
2025-05-04 - 2026-06-07

